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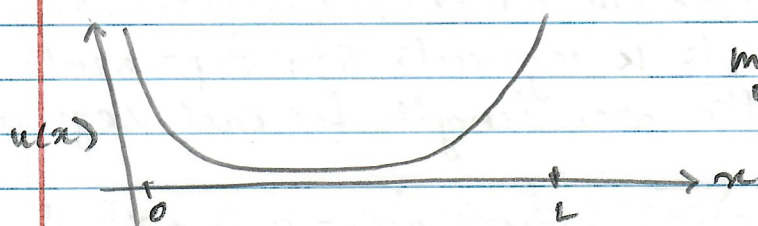
CSE 393P: Computational & Variational Methods for Inverse Problems

10 March 2025

Assignment 2

1. Derivation of the optimization problem

$u: [0,1] \rightarrow \mathbb{R}$ maps the height of the chain



$$\min_u E(u) \quad \text{s.t. } L(u) = L_0 \text{ \& } u(0) = u(1) = 0$$

Create an unconstrained optimization problem w/ quadratic length penalty:

$$\min_u E(u) + \alpha (L(u) - L_0)^2$$

$$\text{s.t. } u(0) = u(1) = 0$$

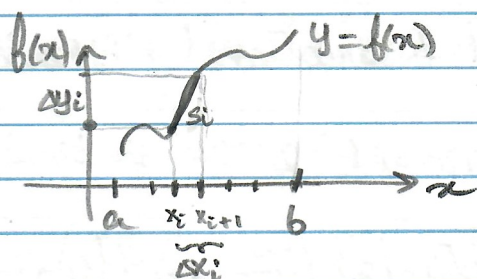
for some $\alpha > 0$
(Note $\lim_{\alpha \rightarrow \infty}$, the problem

becomes the same as the constrained optimization, but is ill-conditioned)

a) Show that the length of the chain $L(u)$ is:

$$L(u) = \int_0^1 \sqrt{1 + (u')^2} \, dx \quad \text{where } u' := \frac{du}{dx}$$

Consider a curve $y = f(x)$ on a domain $[a, b]$: 2



let $L :=$ length of curve on $[a, b]$.

$L = \sum_{i=1}^k s_i$ where we divide the domain $[a, b]$ into k segments & s_i represents the arc length for each segment.

$$s_i \approx \sqrt{\Delta x_i^2 + \Delta y_i^2} \quad \text{where } \Delta x_i = x_{i+1} - x_i \text{ \& } \Delta y_i = y_{i+1} - y_i$$

For small intervals:

$$\lim_{k \rightarrow \infty} s_i = \sqrt{dx_i^2 + dy_i^2} \quad \text{or } ds = \sqrt{dx^2 + dy^2}$$

$$\Rightarrow ds = dx \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$L = \lim_{k \rightarrow \infty} \sum_{i=1}^k ds = \int_a^b ds = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx.$$

for the given hanging chain problem, $x \in [0, 1]$ or $a=0$ & $b=1$. Additionally, instead of the curve being represented as a function $y(x)$, we have $u(x)$ to represent the vertical height

$$\text{So, } L(u) = \int_0^1 \sqrt{1 + \left(\frac{du}{dx}\right)^2} dx = \int_0^1 \sqrt{1 + (u')^2} dx$$

b) Show that the potential energy $E(u)$ is:

3

$$E(u) = \int_0^1 \rho g u \sqrt{1+(u')^2} \, dx \quad \text{*assume density is constant}$$

where ρ is the linear density (mass per unit length) & g is the gravitational acceleration constant.

For a small section/element of the chain, the potential energy

(1) $dE = (\text{mass of element}) g (\text{height of element})$
 $= (\rho dm) g u \quad \approx \text{constant across element due to taking a small section.}$

(2) $dm = \rho \, ds$ where ds represents the differential arc length of the chain.

with (1), (2):

$$dE = (\rho ds) g u$$

$$E(u) = \int_{\text{over all chain}} dE = \int_{\text{over all of chain's arc}} \rho ds g u = \int_0^1 \rho g u (\sqrt{1+(u')^2}) \, dx$$